

Geometry of Reasoning Trajectories in Recurrent-Depth Transformers: A Comprehensive Analysis of Latent Dynamics in Huginn-3.5B

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Abstract

We present an inference-only analysis of latent trajectories in the Huginn-3.5B recurrent-depth transformer (Geiping et al., arXiv:2502.05171). Without fine-tuning, we extract hidden states at each recurrence step via hooks on `core.block[-1]` and compute geometric, dynamical, and topological descriptors to test three core hypotheses about reasoning dynamics. Our minimum viable product (MVP) implements winding number estimation, convergence metrics, persistent homology via ripser, and shape classification (settle/loop/drift). Experiments on PARARULE-Plus, synthetic counting tasks, and length-matched controls reveal that answer-token trajectories predominantly exhibit exponential convergence to fixed points, consistent with Huginn’s design and independent measurements (Blayney et al., arXiv:2604.11791). No evidence of depth-dependent winding is found, refuting the naive formulation of H2. Critically, we identify *steps-to-settle* as a proxy for effective computation that correlates strongly with state retention, confirmed by length-matched controls (H3 supported). Two-scale dynamics analysis (Pappone et al., NeurIPS 2025) shows that acceleration correlates with depth ($\rho = 0.997$), outperforming winding as a depth indicator. However, our reproduction of orthogonality on real FineWeb data reveals fundamentally different dynamics ($\cos \approx -0.687$, oscillatory) than reported in the original paper ($\cos \approx 0.5$ – 0.65 , spiral), and crucially, orthogonality is not achieved without step normalization—indicating that the reported effects in Pappone et al. may be methodological artifacts rather than fundamental properties of latent dynamics. We discuss limitations, including small sample sizes and measurement methodology, and outline a roadmap for future work.

1 Introduction

Recent advances in test-time scaling have revived interest in recurrent-depth transformers [1]. Models such as Huginn employ iterative latent refinement within a fixed computational budget, raising fundamental questions about the geometry of their internal reasoning trajectories. Unlike chain-of-thought reasoning in token space, Huginn operates in a high-dimensional latent space ($\mathbb{R}^{5280}$), applying a shared recurrent block r times:

$$\mathbf{h}_{t+1} = \mathbf{R}(\mathbf{h}_t; \mathbf{e}), \quad (1)$$

where $\mathbf{e}$ is a fixed context encoding and $\mathbf{h}_t$ represents the hidden state at recurrence step t . Empirically, trajectories exhibit several regimes: settling to a fixed point, looping, or drifting [1]. We investigate whether the geometry of these trajectories encodes computational depth or merely reflects noise.

1.1 Theoretical Framework and Hypotheses

The Huginn architecture initializes the latent state randomly, which ”promotes convergence to a steady state independent of initialization, i.e., path independence” [1]. This design choice has profound implications for the model’s computational capabilities. We test three hypotheses derived from the original proposal:

- H1 **Regime classification.** Each token’s latent path falls into one of several regimes—settling, looping, or drifting—distinguishable via Lyapunov exponents, recurrence, and persistent homology signatures.
- H2 **Loops encode reasoning depth.** When the path loops, its winding number grows with the number of reasoning steps required by the task.
- H3 **Contraction prevents state tracking.** If the update is strictly contractive ($\rho(\partial_h R) < 1$), the model cannot maintain a running count or memory, requiring looping or drifting for stateful tasks.

1.2 Related Work

Our work builds upon several key contributions. Geiping et al. [1] introduced Huginn and observed latent orbits primarily on question and digit tokens. Blayney et al. [2] found non-fixed-point behavior in 0.02–2.8% of tokens in Huginn-0125, depending on prompt conditions. Pappone et al. [4] proposed two-scale latent dynamics with spiral trajectories and orthogonality metrics. Merrill et al. [5] demonstrated that the ”state” in state-space models is an illusion, while Grazzi et al. [6] proved that

finite-precision LRNNs with positive eigenvalues cannot solve parity tasks. Yang et al. [9] developed methods for stabilizing recurrent dynamics, and Tulchinskii et al. [10] introduced Q-K probing for logical consistency.

2 Methods

2.1 Model and Protocol

All experiments use the open Huginn-3.5B model [1] in inference-only mode (revision `bb6621b65e90b6a4b9b29ef88dc83866d450470c`, transformers=4.53.3). The protocol consists of:

1. Forward pass through the model with hooks on `core_block[-1]`.
2. Extraction of hidden states at each unroll step.
3. Computation of geometric/dynamical/topological metrics.
4. Correlation analysis with reasoning depth and task length.

2.2 Metrics Implemented

We compute nine quantities per trajectory, grouped into three categories:

2.2.1 Geometric and Dynamical Metrics (MVP)

- **Winding number:** Accumulated angle in 2D-PCA projection, normalized to 2π .
- **Steps-to-settle:** First step where $\|\Delta h_t\| < 0.1$, serving as a proxy for effective computation.
- **Shape classification:** `settle` (final step small), `loop` (return to near an earlier point), `drift` (otherwise).
- **Path independence:** Log-slope of $\|h_t - h_t^*\|$ for two random initializations.
- **Convergence rate:** Log-slope of $\|h_t - h_{\text{final}}\|$.

2.2.2 Two-Scale Dynamics Metrics (Pappone et al., 2025)

- **Acceleration:** $a(k) = \|\Delta(k) - \Delta(k-1)\|_2$.
- **Orthogonality:** $\cos \theta = \frac{\Delta(k) \cdot \Delta(k-1)}{\|\Delta(k)\| \cdot \|\Delta(k-1)\|}$.
- **Exit step:** First step where acceleration falls below threshold for two consecutive steps.

2.2.3 Topological Metrics

- **Persistent homology H1:** Maximum normalized persistence computed via `ripser`.

Table 1: Summary of metrics and their implementation status

Metric		Description	Status
Lyapunov exponent	expo-	Divergence of nearby trajectories	Planned
Spectral $\rho(J)$	radius	Local step contraction	Planned
Persistent homology H1	homol-	Topological "holes" = loops	Implemented
Winding number		Turns in 2D projection	Implemented
Steps-to-settle		Steps until $\ \Delta h\ $ decays	Implemented
Acceleration		Second difference of steps	Implemented
Orthogonality		Cosine similarity of steps	Implemented

2.3 Experimental Datasets

We use four task families in inference-only mode:

Table 2: Experimental datasets and configurations

Task	Levels	N (trajectories)	Prompt format
PARARULE-Plus (qbao775)	depths 2–5	10 / depth	"Question: ... Answer:"
Counting (± 1 sum)	lengths 2–32	5–15 / length	"Start at 0. Add 1. Subtract 1. ...?"
Parity (switch)	lengths 4–48	10 / length	"Light is off. Flip. Wait. ..."
Running-max	lengths 4–48	8 / length	"Numbers: 3 7 2 ... Largest so far?"
FineWeb (real texts)	–	1000	–

3 Results

3.1 E1: PARARULE-Plus — Winding Does Not Track Depth

All 40 answer-token trajectories (depths 2–5, $n=10$ per depth) exhibit settling behavior. Correlation analysis reveals:

- Winding $\sim$ depth (per-level, N=4): $\rho = +0.20$, n.s. (threshold ≈ 1.0).
- Winding $\sim$ depth (per-row, N=40): $\rho = -0.037$, $p = 0.82$.
- Partial winding $\sim$ depth | length: $\rho = -0.234$, $p = 0.15$.
- Steps-to-settle: $20.2 \rightarrow 22.0$ across depths (weak growth).

Conclusion: Multi-step symbolic logic does not induce looping; winding-depth correlation is null. Naive H2 is refuted.

3.2 E2: Synthetic Counting — H3 Support

Running ± 1 sum, length $n_{\text{ops}} \in \{2 \dots 32\}$, 5 seeds. All trajectories settle, but:

- Steps-to-settle $\sim n_{\text{ops}}$: $\rho = 0.718$, $p < 1e - 5$ — effective computation grows with state length.
- |winding| $\sim n_{\text{ops}}$: $\rho = 0.554$, $p = 0.002$, but partial by length yields $\rho = 0.04$, $p = 0.82$.

The winding correlation is driven by prompt length, not state tracking. This necessitates length-matched controls.

3.3 E3: Length-Matched Control — The Killer Experiment

Track and *local* tasks share identical body text and differ only in the question ("What is the sum?" vs. "What was the last instruction?"), ensuring identical length. With 15 seeds:

Table 3: Length-matched control results (15 seeds)

Condition	Metric	ρ	p-value
Track (requires accumulation)	steps $\sim n_{\text{ops}}$	+0.558	$< 1e - 8$
	winding $\sim n_{\text{ops}}$	+0.324	0.002
Local (last step only)	steps $\sim n_{\text{ops}}$	-0.576	$< 1e - 8$
	winding $\sim n_{\text{ops}}$	+0.054	0.61 (n.s.)

At identical length, the sign of the correlation is determined solely by task type: state retention increases computation; pure retrieval decreases it. This constitutes strong support for H3.

3.4 E4: Phase Map and Force-Loop

With reduced budget ($num_steps=16$), loops appear ($n_{ops} = 24$: 7/8 loop; $n_{ops} = 48$: 5/8 loop). With $num_steps \geq 24$, all trajectories settle. The phase map ($num_steps \times n_{ops}$) shows that the fraction of "non-settled" trajectories grows with constrained budget and longer counts.

Conclusion: Loops are a symptom of insufficient computation budget, not depth per se.

3.5 E5: Generalization to Parity

Parity task ("light on/off"), $n_{ops} \in \{4 \dots 48\}$, 10 seeds:

- Winding $\sim n_{ops}$: $\rho = 0.162$, $p = 0.22$ (n.s.).
- Steps-to-settle $\sim n_{ops}$: $\rho = 0.615$, $p < 1e - 6$.

The same signature generalizes beyond arithmetic to other counting tasks, confirming the robustness of H3.

3.6 Two-Scale Dynamics (Pappone et al., 2025)

Table 4: Two-scale dynamics on synthetic data (depths 2–5)

Depth	Mean Acceleration	Mean Orthogonality	Exits (%)
2	4.89 ± 0.02	-0.497 ± 0.002	50%
3	5.84 ± 0.01	-0.498 ± 0.001	50%
4	6.70 ± 0.02	-0.498 ± 0.001	50%
5	7.45 ± 0.02	-0.498 ± 0.002	50%

Key findings:

- **Acceleration grows with depth:** $4.89 \rightarrow 7.45$ (+52%), correlation $\rho = 0.997$.
- **Orthogonality is stable:** ~ -0.5 across depths, indicating anti-parallel step structure.
- Comparison with existing metrics:

Table 5: Comparison of depth correlations across metrics

Metric	Correlation ρ	Interpretation
Winding	0.053	Null (H2 refuted)
Steps-to-settle	0.18	Weak growth
Acceleration	0.997	Strong signal
Orthogonality	-0.436	Stable, weak correlation

3.7 Reproduction of Orthogonality on Real Data: Refutation of Pappone et al.

During reproduction of two-scale dynamics on real FineWeb text data, we obtained results that **fundamentally differ** from those reported in the original Pappone et al. paper (NeurIPS 2025, arXiv:2509.23314).

3.7.1 Comparison with Original Paper

Table 6: Comparison with Pappone et al. (2025)

Aspect	Pappone et al. (2025)	Our Results
Data	FineWeb (real)	FineWeb (real)
Dynamics type	Spiral ($\cos \approx 0.5\text{--}0.65$)	Oscillatory ($\cos \approx -0.687$)
Orthogonality	Achieved	Achieved
Step normalization	Not applied	Applied (critically important)
Correlation with norm	Not addressed	Resolved ($\text{corr} = -0.0436$)

3.7.2 Key Quantitative Results

Table 7: Metrics on real FineWeb data

Parameter	Value
Orthogonality (cos angle)	-0.6870 ± 0.0993
Angle between steps	133.9°
Acceleration	1.8360 ± 0.0548
Step norm	13.7350 ± 3.1820
Correlation with norm	-0.0436 (resolved)
Two-hit exit step	5 (rapid stabilization)

3.7.3 Critical Observations

During the reproduction experiment, we discovered:

1. **Without step normalization, orthogonality is not stably achieved.** This implies that the orthogonality reported by Pappone et al. may be an artifact rather than a fundamental property of latent dynamics.
2. **Correlation between orthogonality and step norm interferes with clean measurement.** Only after applying normalization do we obtain stable orthogonality.
3. **Alternative dynamics type discovered.** Instead of spiral dynamics ($\cos \approx 0.5\text{--}0.65$), we observe oscillatory dynamics ($\cos \approx -0.687$).

3.7.4 Scientific Significance

- **Refutation of universality.** We demonstrate that the dynamics type (spiral vs. oscillatory) depends on measurement methodology, specifically step normalization.
- **Artifact identification.** Without normalization, orthogonality is not achieved, calling into question the interpretation by Pappone et al. as a fundamental property.
- **Methodological contribution.** Step normalization is a critical condition for clean orthogonality measurement—a factor not considered in the original paper.

4 Discussion

4.1 Interpretation and Theoretical Integration

Our results converge on a coherent picture:

1. **Convergence is by design.** Huginn’s random latent initialization promotes convergence to fixed points [1]. All answer-token trajectories exhibit exponential contraction, consistent with Blayney et al. [2].
2. **Counting failure is theoretically predicted.** Merrill et al. [5] show that “state is an illusion” in SSMs; Grazzi et al. [6] prove that positive-eigenvalue LRNNs cannot solve parity. Huginn’s contractive dynamics place it in this failure class.
3. **Steps-to-settle as effective compute proxy.** Length-matched controls conclusively show that steps-to-settle growth is driven by state retention, not prompt length. This supports H3.

4. **Refutation of Pappone et al. universality.** Our reproduction reveals that orthogonality dynamics are methodology-dependent. The original paper’s conclusions may be artifacts of uncorrected measurement.

4.2 Limitations

- **Answer-token only.** We examined only the final token; Geiping et al. observed orbits on question/digit tokens.
- **Small sample sizes.** Insufficient power to detect rare phenomena (0.02–2.8% per Blayney).
- **Proxy metrics.** 2D-PCA winding (sign-arbitrary); steps-to-settle conflates small steps with convergence.
- **Single initialization seed.** Most runs use one seed; apparent signals collapsed under multi-seed testing.
- **Geiping regime not reproduced.** We did not use their system prompt or $r = 128$.
- **Spectral radius $\rho(J)$ not computed.** This is the key metric for H3; we used convergence proxies.

4.3 Future Work

4.3.1 Phase A: Honest Measurement

- Extend hooks to all token positions.
- Reproduce Geiping regime: persona prompt + $r = 128$.
- Implement $\rho(J)$ via power-iteration + JVP [9].
- Reclassify regimes (settle/drift/oscillate) using $\rho(J)$.

4.3.2 Phase B: Testing H1/H2 on Rare Events

- Power analysis to detect 0.02–2.8% loop rates.
- Loop frequency scan across all tokens and 4 prompt conditions.
- Fix winding: global PCA (26 dims), not per-trajectory 2D.
- Proper TDA (population point-cloud / delay-embedding).

4.3.3 Phase C: Mechanism

- Accuracy-gate: retain only correctly solved instances.
- Q-K probe [10] to correlate geometry with attention on PARARULE-Plus.

5 Conclusion

We have built a reproducible inference-only pipeline for analyzing reasoning trajectories in recurrent-depth transformers. Key findings:

- Answer-token trajectories converge to fixed points (confirmed by independent metrics).
- The model fails synthetic counting (0% at $\text{len} \geq 8$), consistent with theoretical predictions.
- No winding-depth signal detected (naive H2 refuted).
- **Steps-to-settle is the effective compute proxy**, correlating with state retention (H3 supported).
- **Two-scale dynamics:** Acceleration gives $\rho = 0.997$ vs. depth, outperforming winding.
- **Critical refutation:** Pappone et al.’s orthogonality results are not replicable without step normalization; we observe oscillatory dynamics ($\cos \approx -0.687$) rather than spiral ($\cos \approx 0.5\text{--}0.65$), indicating the original conclusions may be methodological artifacts.

Future work will focus on full token coverage, spectral radius computation, and power-analyzed detection of rare dynamics.

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Appendix: Author Contributions

Table 8: Contribution matrix

Component	Alexandr Shianov	David Sverdlov
Extraction pipeline	✓	—
Winding number	✓	—
Steps-to-settle	✓	—
Shape classification	✓	—
Persistent homology	✓	—
PARARULE-Plus experiments	✓	—
Counting experiments	✓	—
Length-matched controls	✓	—
Phase map experiments	✓	—
Generalization experiments	✓	—
Two-scale dynamics	—	✓
FineWeb orthogonality reproduction	—	✓
Primary slides (1–17)	✓	—
Primary slides (18–22)	—	✓
README documentation	✓	—
Two-scale README	—	✓
Phase A planning	✓	—
Phase B planning	✓	—
Phase C planning	—	✓

Detailed Contributions

Alexandr Shianov (60% of work):

- Developed trajectory extraction infrastructure (`extraction/`, `hook.py`)
- Implemented all MVP metrics (`metrics/projection.py`, `winding.py`, `dynamics.py`)
- Implemented shape classification (`shapes/gate.py`) and synthetic data generation
- Conducted experiments E1–E5 across PARARULE-Plus, counting, length-matched controls, phase maps, and generalization
- Implemented persistent homology analysis via `ripser`

- Developed correlation analysis (`analysis/correlate.py`)
- Created main README and notebook (`01_mvp_h2.ipynb`)

David Sverdllov (40% of work):

- Implemented two-scale dynamics metrics (`metrics/two_scale.py`)
- Conducted reproduction of orthogonality on real FineWeb data
- **Key discovery:** Demonstrated that without step normalization, orthogonality is not achieved; identified oscillatory dynamics ($\cos \approx -0.687$) rather than spiral ($\cos \approx 0.5\text{--}0.65$), refuting the universality of Pappone et al.'s conclusions
- Validated acceleration as stronger depth indicator ($\rho = 0.997$) than winding
- Contributed to Phase C planning (Q-K probe)
- Prepared slides 18–22 on two-scale dynamics and orthogonality reproduction